

# INTRODUCTION

## Code First Girls

### *Project Assignment*

| GROUP SIX - “MyMoviePal”                                                                                 |
|----------------------------------------------------------------------------------------------------------|
| Katie John<br>Laura Dolan<br>Mayte Bujalance Casas<br>Omoeffe Osayi<br>Palvi Williams<br>Tayibah Hussain |

#### **Picture this.**

You’ve just finished a long shift at work on a Friday, the weekend is ahead of you and you have no plans other than pure relaxation. Dinner has been made and you would like nothing more other than to watch a film before retiring for the evening. Only, there are thousands upon thousands of films readily available to us through streaming platforms and sometimes, deciding on the perfect film can turn an evening of relaxation into one of unbridled frustration.

Group Six has decided on a solution to combat this very issue: a program, coded by us, named ‘**MyMoviePal**’. The application would function as a movie recommendation app, where the program would prompt the user to answer a series of questions personal to the user, in order to tailor the correct movie based on a multitude of data.

Whilst there are already applications available, such as streaming websites (Netflix, Prime Video etc) which already incorporate a ‘movie list’ function, our program aims to include films from multiple streaming sites, all of which will detail if they are free to watch or included in a prepaid subscription, allowing the user to feast on a broad selection of movies. This application will be easy to use, seamless enough to be integrated into watch parties with friends whilst also ensuring that each recommendation qualifies against the criteria; provided by the user-answered questions.

The question of how Group Six will manage to pull off this idea at a professional level can easily be answered. From a technical perspective, MyMoviePal will be built as a Flask-based application which will siphon information from a SQL database used to store data such as user accounts, movie ratings, friend/follower activity and the user’s own personal watch history. An idea discussed was that MyMoviePal will pull information from an external API, such as TMDb, to retrieve data such as the cast list, movie titles and release dates, possibly even including the run time of each movie if this is considered a user-based preference.

THE PROBLEM?

## **What problem does ‘MyMoviePal’ fix?**

Firstly, ‘MyMoviePal’ answers the age-old question: **“What should we put on the telly?”**

From personal experience, my group members and I have been to many movie nights or settled down after a long day of work, networking or just a generally bad day and needed something to unwind to. If we look back at around twenty years ago, we did not ponder over a problem such as this because the threat of choice didn’t plague us. There weren’t a number of streaming sites available; you’d go to the local Blockbuster or HMV to rent or buy a film from their small selection and that was that. Now, we face the issue of quantity. Thousands of movies are readily available through an online click, movies dating as far back as the black-and-white silent film days and works from actors since the beginning of their careers to present day.

So, how exactly do we, the creative minds behind ‘MyMoviePal’, combat this?

MyMoviePal is a personalised program which tailors movie recommendations unique to each user. The idea is that MyMoviePal will function like a very early AI chatbot, asking simple questions to the user to narrow down the database of movies on offer. Each movie will be tagged by year of release, actors, genre, runtime, star rating and overall pick rate so that when the user fills in the criteria, MyMoviePal will recommend only the best matches.

Since each recommendation is unique to the user, it provides an interesting assortment of movies for users who partake in watch parties or movie nights with friends. Every user is different, where there may be similar choices between friends, MyMoviePal will ensure that each user receives recommendations tailored to their personal taste above all.

Not only does MyMoviePal solve that quandary, but by analysing a wide range of streaming sites, the user is able to save money on subscriptions by reducing the amount of sites they are subscribed to. Already paying for Prime Video, Netflix, Disney Plus etc? MyMoviePal will tell you where the film you want is streaming, making it easier for the user to budget their monthly TV subscriptions.

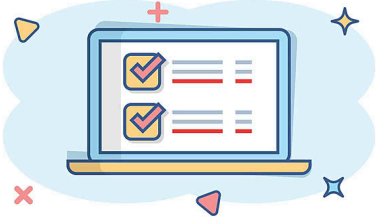
Perhaps you have already decided on a movie, a friend from work has recommended a movie they’ve watched and it ticked all the boxes, but you’ve forgotten the title. No worries there! MyMoviePal’s extensive filtering system with multiple tags in different categories allows you to fill in the information you already know and it will find the movie you want. No more spending hours searching through IMDb on an actor’s filmography, MyMoviePal covers all ground for you. It’s easy to use, wildly accessible and will function as an AI bot does with customised greetings and seamless features allowing you to always look forward to a movie night.

# FEATURES

## What are the main features of your application?

Upon launch, 'MyMoviePal' will have the features listed below. However limited, the upcoming features section details our planned additions in a later update.

*Thank you Mayte for creating the features list used below.*

| CURRENT FEATURES                                                                                                                                                                        |                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>User questionnaire</b> to gather data on movie preferences.</p>                                  | <p><b>Recommendation engine</b> based on user responses.</p>                                                                                                                    |
| <p><b>Detailed movie information display.</b></p>                                                     | <p><b>Search and filtering</b> functionality.</p>                                                                                                                              |
| UPCOMING FEATURES <i>(pending future updates)</i>                                                                                                                                       |                                                                                                                                                                                                                                                                   |
| Integration with <b>movie and streaming APIs</b> .                                                                                                                                      | User accounts and <b>saved favourites</b> .                                                                                                                                                                                                                       |
| <b>Watchlist</b> functionality.                                                                                                                                                         | Friend and group <b>recommendations</b> .                                                                                                                                                                                                                         |
| <b>Recommendation</b> history.                                                                                                                                                          | Movie night / <b>watch party</b> mode.                                                                                                                                                                                                                            |
| <p><b>Advanced filtering</b> by:</p> <ul style="list-style-type: none"><li>➤ Genre</li><li>➤ Release year</li><li>➤ Runtime</li><li>➤ Age rating</li><li>➤ Streaming provider</li></ul> | <p><b>'PalRewind'</b>: A look back on your MyMoviePal activity over the previous year, similar to Spotify Wrapped.</p> <p>Each month will return your most watched activity, recommending said movies to you for another watch based on rating and user mood.</p> |

DIAGRAM

## How does ‘MyMoviePal’ work?

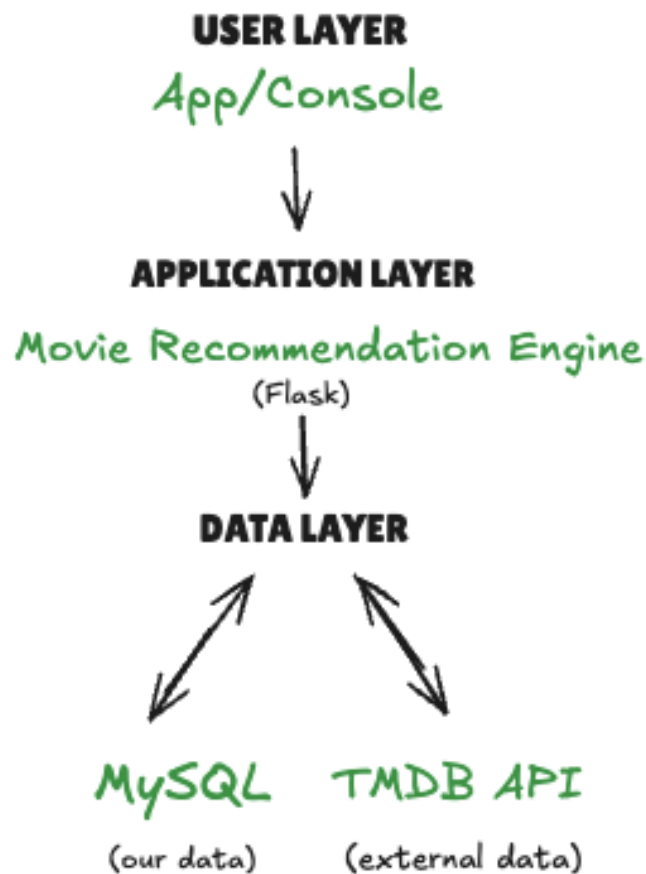
The diagram below provides a simplified overview of how ‘MyMoviePal’ is structured across three layers:

**User Layer** - The user interacts with MyMoviePal through the app/console, inputting their preferences and receiving tailored movie recommendations.

**Application Layer** - Flask serves as the engine of the application, processing user inputs and handling all the routes, logic and functionality behind the scenes.

**Data Layer** - Flask communicates with two key data sources to power MyMoviePal’s recommendations.

- **MySQL** stores all of our own data (user accounts, watchlists, ratings and follows)
- **TMDB API** provides live external film data including movie details, actor information and recommendations



# TEAM APPROACH

## How will the team approach the project?

In order to complete our project at the highest standard attainable, each of us will contribute our own ideas and code to a specific branch within our repository. For more detail on personal involvements, see the table below.

|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Katie</b></p> <p><b>Katie</b> will handle the recommendation filter and tagging, creating lists and dictionaries to tag specific movies per genre (using if/else statements), working closely with <b>Efe's</b> SQL database and <b>Laura's</b> API integration to ensure that <b>Palvi's</b> front end runs smoothly.</p>                                                   | <p><b>Laura</b></p> <p><b>Laura</b> will focus on integrating the TMDB API into the project, providing MyMoviePal with the data needed for the app to call upon specific movies tailored to user taste. She will work closely with <b>Katie's</b> recommendation filters and <b>Efe's</b> SQL database to ensure all data provided is correct.</p>         |
| <p><b>Mayte</b></p> <p><b>Mayte</b> will handle the Flask app route and core setup of the app, giving our project the backend structure it needs. Working closely with <b>Laura's</b> API integration and <b>Palvi's</b> front end development to create a smooth running, easy to use application.</p>                                                                            | <p><b>Palvi</b></p> <p><b>Palvi</b> will take the responsibility of creating the front-end of the application, guiding the user through the series of questioning, mirroring a very early prototype of an AI chatbot. She will work with <b>Katie</b> and <b>Mayte</b>, who will provide the spine and filtered content, tailored by user taste.</p>       |
| <p><b>Omoefe</b></p> <p><b>Efe</b> has chosen to create the MySQL database, taking information from the TMDB API to store user data, watchlists, and providing the necessary information for our 'PalRewind' feature. <b>Laura</b> and <b>Katie</b> will assist on this task in order for the flow from API to database to content filtering will run as smoothly as possible.</p> | <p><b>Tayibah</b></p> <p><b>Tayibah</b> has the most important task. She will be responsible for testing and debugging the project, working with everyone to minimise the amount of mistakes and to ensure all sections are coded perfectly, recommending easier alternatives if there is a section of messy code which could inhibit user experience.</p> |

Whilst the colour-coded roles act as guidelines for now, our team has decided that we will work closely together and assist where needed. Every person has an important role to ensure that MyMoviePal will run smoothly and each of us will take the time to discuss, plan and go through continuous periods of trial and error to promote user satisfaction but to also satisfy ourselves as the developers on a project that we're proud of.